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Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

DESIGN AND TECHNOLOGY**4053/1**

Paper 1 Design and Technology Theory

3 hours**SPECIMEN PAPER**

Additional materials:

Answer booklet, Calculator,
 Coloured pencils, 2 × (A3) Drawing paper,
 Standard drawing equipment.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on the separate answer booklet used.

Section A [40 marks]Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B [60 marks]Answer **four** questions, **one** from each of the **four** options given.

At the end of the examination, fasten the separate answer booklet and drawing papers securely to the question paper.

INFORMATION FOR CANDIDATES

Marks are given in brackets [] at the end of **each** question or **part** question.

You are advised to spend not more than **45 minutes** on**Section A** and not more than **2 hours 15 minutes** on **Section B**.**FOR EXAMINER'S USE**

A1	
A2	
A3	
A4	
A5	
A6	
A7	
A8	
B9	
B10	
B11	
B12	
B13	
B14	
B15	
B16	
TOTAL	

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SECTION A (40 Marks)

*You are advised to spend **not** more than **40 minutes** in this section.*

Answer all the questions in the spaces provided.

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- 1 (a) Figure 1** shows a Personal Protective Equipment (PPE) chart in a workshop.

Complete the PPE chart by filling in the function of each sign.

	Sign	Function
(i)		
(ii)		
(iii)		

Fig. 1

[3]

- (b)** State any **two** reasons for displaying signages in a workshop.

[2]

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- 2 (a) Give any **three** advantages of registering a company in Zimbabwe.

[3]

- (b) Name the document that is issued out by the Deeds Office of Zimbabwe, after a company is registered.

[1]

- (c) Identify the Intellectual Property Right (**I.P.R.**) that protects a company logo.

[1]

- 3 (a) Identify any **two** types of drilling machines used in a workshop.

[2]

- (b) Explain any **three** safety precautions observed when using a power drill.

[3]

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- 4 **Figure 4(a)** shows a diagram of a flag (A) on post (B).

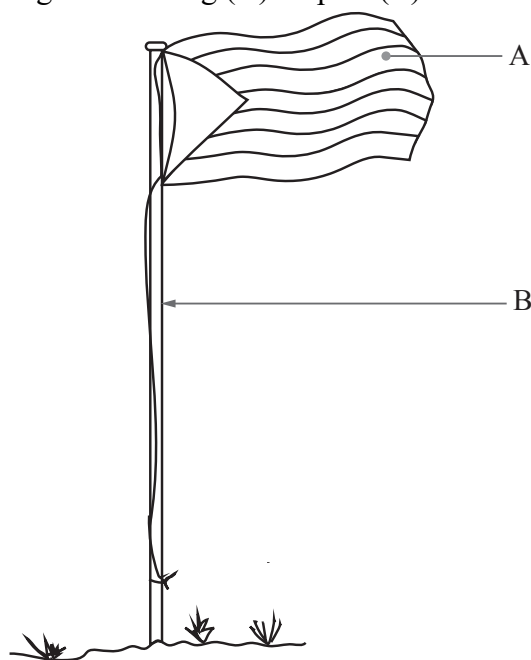


Fig. 4(a)

- (a) Identify any suitable materials for making parts **A** and **B**.

A _____

B _____ [2]

- (b) (i) Describe how part **B** can be protected from weather elements.

 _____ [1]

- (ii) Suggest any **two** reasons why the material you have suggested to make part **A**, is suitable.

 _____ [2]

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- 5 (a) State any **two** distinct current features of the Zimbabwean indigenous architecture.

[2]

- (b) **Figure 5(b)** shows a diagram of a wooden coffee table that was made in a school workshop.

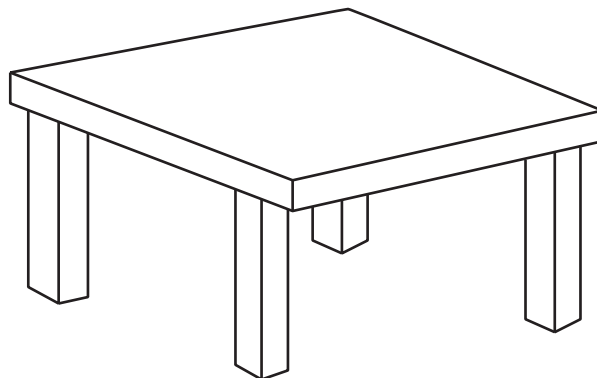


Fig. 5(b)

Suggest any **three** modifications that can be done to the coffee table to make it more stable and more user friendly.

[3]

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6 **Figure 6** shows an electrical circuit.

Name the components labelled A – E.

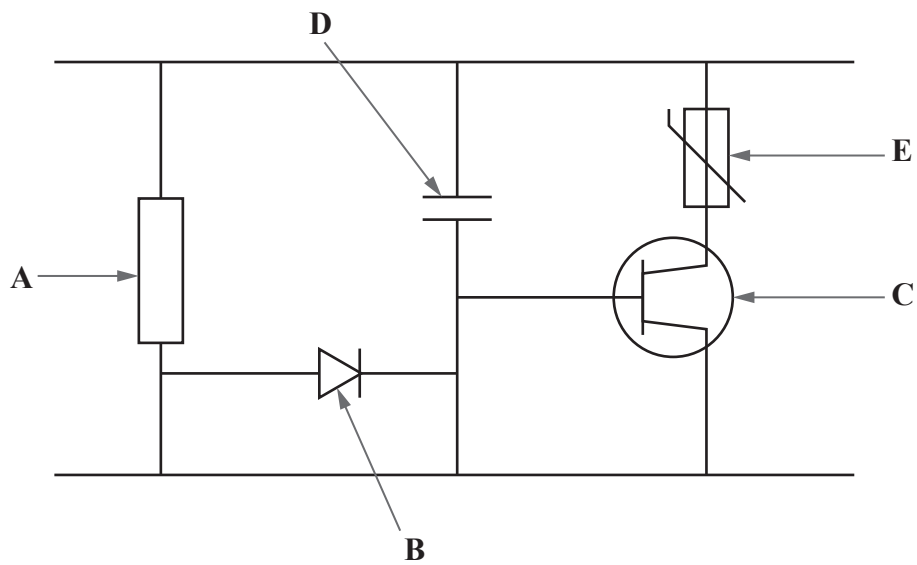


Fig. 6

A _____

B _____

C _____

D _____

E _____

[5]

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- 7 **Figure 7** shows part of a machine which lifts grain bags from the ground onto a scotch cart.

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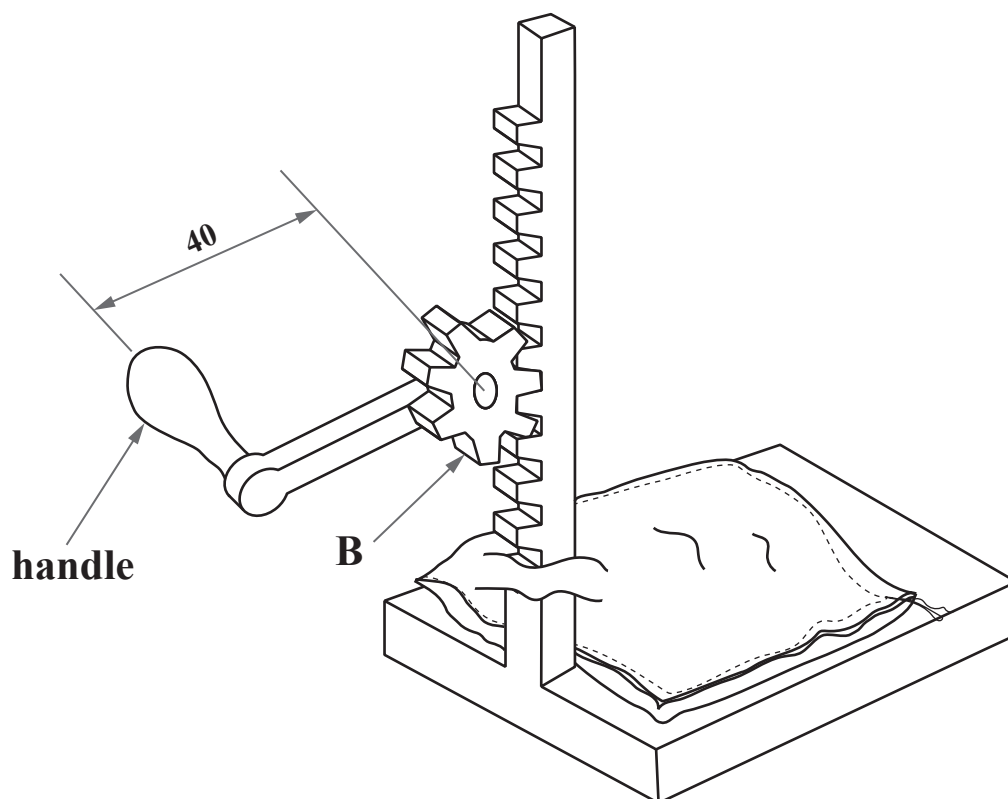


Fig. 7

- (a) Name the type of gears the machine is using.

[1]

- (b) If the handle is **40 mm** long and the diameter of pinion **B** is **16 mm**, calculate the velocity ratio of the gear system.

[2]

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- (c) Determine the effort used if the weight of the bag is **800 N**.

_____ [2]

- 8** (a) Identify any **two** manufacturing systems.

_____ [2]

- (b) Give any **three** components of a manufacturing system.

_____ [3]

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SECTION B (60 Marks)

*You are advised to spend **not** more than **2 hours 15 minutes** on this section.*

*Answer **four** questions from this section in the separate answer booklet provided.*

*Choose **one** question from each of the options, Resistant materials, Systems and Control, Engineering Science and Manufacturing.*

Resistant materials

- 9 (a) Give any **two** differences between thermoforming and thermosetting plastics. [4]
- (b) Describe how a test to determine rate of corrosion in resistant materials can be done. [4]
- (c) Describe the process of *plastic dip* coating. [5]
- (d) **Figure 9** shows a symbol found on packaging of locally produced products.

**Fig. 9**

- (i) What is the meaning of the abbreviation **S.A.Z**? [1]
- (ii) Suggest a reason for including the symbol on products packaging. [1]

[15]

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- 10**
- (a) Identify any **one** material used for preventing moisture penetration into a building. [1]
 - (b) Suggest any position on a building, where the material mentioned in (a) can be placed, to stop moisture penetration. [1]
 - (c) Define wood *seasoning*. [2]
 - (d) Calculate the percentage of moisture content in a stack of partially seasoned timber, given that the weight of wet timber is **45 g** and that of dry timber is **36 g**. [4]
 - (e) What is equilibrium moisture content? [1]
 - (f) Identify any **three** types of weaving. [3]
 - (g) Suggest any **three** items in the local community, that are produced by weaving. [3]
- [15]**

Systems and Control

- 11** A simple bridge is designed to span a river. The bridge resists compressional and tensional forces.
- (a) Explain the difference between the **two** forces. [2]
 - (b) Use a line diagram to illustrate how the bridge structure would resist these forces. [4]
 - (c) Explain any **three** differences between shell and framed structures. [6]
 - (d)
 - (i) Suggest any **two** factors that affect the stability of structures. [2]
 - (ii) State any **one** method of ensuring stability in structures. [1]
- [15]**

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12 (a) Define the following terms as used in electronics:

(i) power, [2]

(ii) voltage. [2]

(b) **Figure 12** shows an electronic circuit with **two** resistors, a lamp and a **6 V** battery.

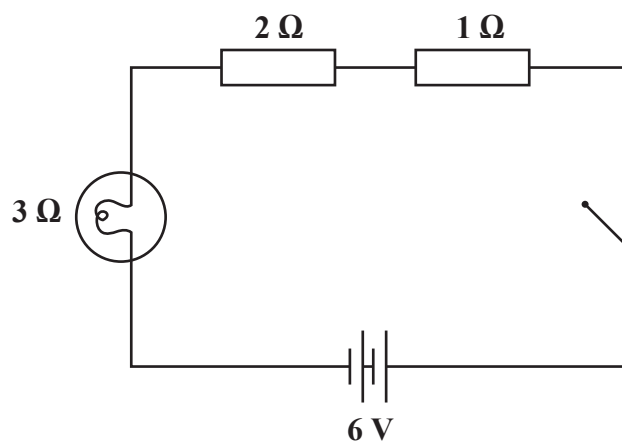


Fig. 12

Calculate:

(i) the current flowing in the circuit, [2]

(ii) the power given out by the lamp. [2]

(c) (i) Name any **two** basic types of capacitors. [2]

(ii) Give any **two** functions of capacitors. [2]

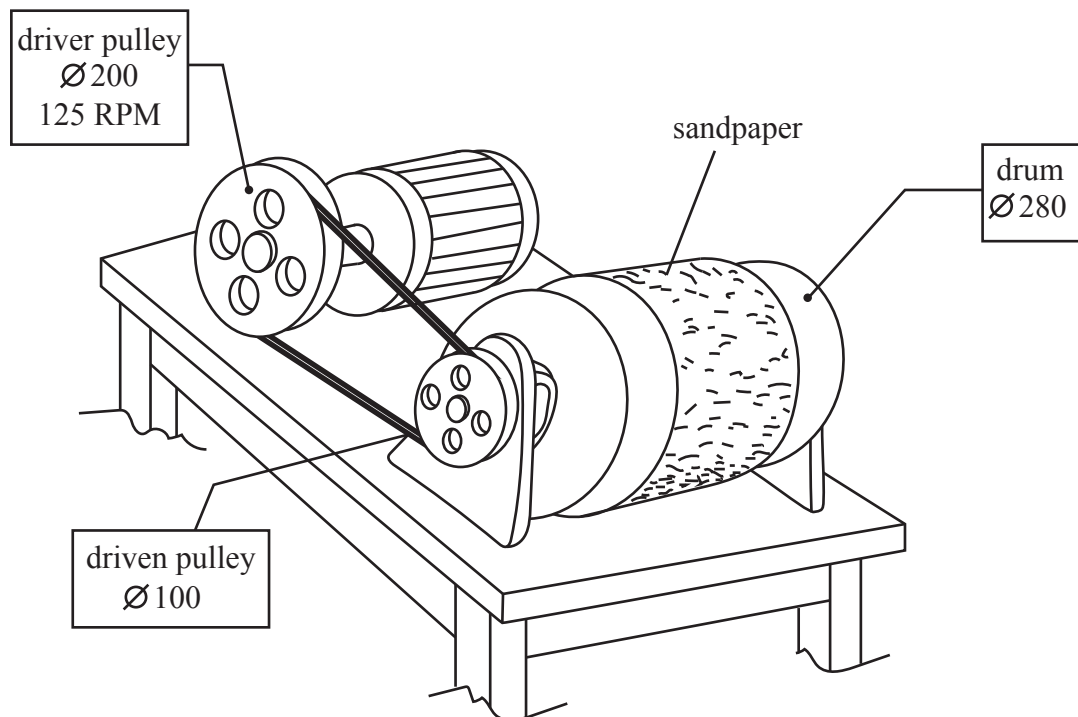
(iii) Name the **three** terminals of a transistor. [3]

[15]

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Engineering Science**13** Figure 13 shows a grinding machine.**Fig. 13**

- (a) Using the information given on the diagram, calculate:
- (i) the velocity ratio of the driver and the driven pulleys, [3]
 - (ii) the speed of the driven pulley, [3]
 - (iii) the spindle speed of the drum in RPM. [3]
 - (iv) the circumference of the drum. [3]

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(b) Suggest how the machine operator can be protected from:

(i) pulleys and belts in motion, [1]

(ii) the revolving drum, [1]

(iii) flying grit from the grinder. [1]

[15]

14 (a) Define the term *moments of force*. [2]

(b) Figure 14(b) shows a spanner turning a nut. A force of **80 N** is applied on the handle of the spanner from the pivot.

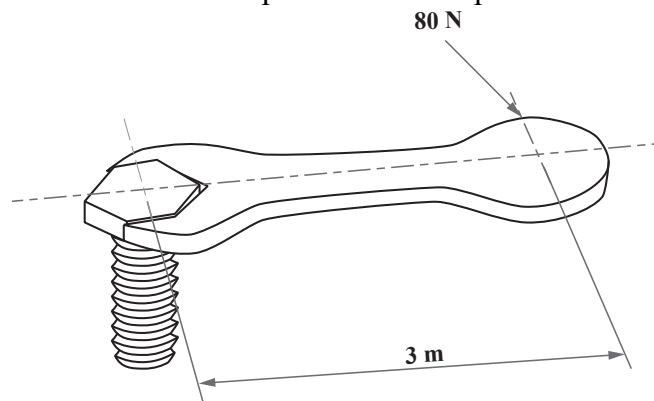


Fig. 14(b)

Calculate the moment of force in this system. [3]

(c) A pivoted beam has **two** point loads which are both **100 N**. **R₁** is **3 m** from the pivot whilst **R₂** is **2 m** away from the pivot on the other side. Each of the **100 N** loads is centrally located between the reaction point and the pivot.

(i) Draw a simple force diagram to illustrate the information given in (c). [6]

(ii) Determine the reactions **R₁** and **R₂**. [4]

[15]

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Manufacturing

- 15** (a) Give any **two** methods of moulding used in manufacturing. [2]
- (b) Identify any **three** products from any **one** of the moulding methods in (a). [3]
- (c) Illustrate using diagrams, **one** of the moulding methods mentioned in (a). [8]
- (d) Suggest any **two** disadvantages of mass production. [2]
- [15]**
- 16** (a) Identify any **two** types of plant layout. [2]
- (b) Suggest any **two** advantages of each of the plant layouts in (a). [4]
- (c) Give any **four** factors that are considered in plant layout. [4]
- (d) Draw a plant layout suitable for the following tools and equipment:
- (i) electrical power equipment,
- (ii) **three** work benches,
- (iii) bench saw. [5]
- [15]**